PADHAKU
Exploration · Class 9 Science



CHAPTER 4

Describing Motion Around Us

NCERT Exercise Solutions

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✨ Try each question yourself first, then check the answer. 💡 answers are samples: write your own.

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🤔 Think It Over

1 **How much distance should we maintain from the truck ahead to avoid a collision if it suddenly applies the brakes?**

💡 SAMPLE ANSWER

We should keep a **safe distance**: at least the distance our vehicle travels during our reaction time plus the distance it needs to stop after braking.

So the gap must be more than our full stopping distance.

2 **Does this distance depend upon the speed with which we are moving?**

✔ ANSWER

Yes. The faster we move, the larger the distance we cover while reacting and while braking.

Braking distance $s = u^2/(2a)$, so doubling the speed makes the braking distance about **four times**.

❓ In-text question · 4.1.2 Distance travelled and displacement

? **Can the total distance travelled and the magnitude of displacement ever be equal?**

✔ ANSWER

Yes, when the object moves in a straight line in **one direction only**, without turning back.

Activity 4.1: Let us analyse

1 **A ball is thrown vertically upwards from O. It moves up straight till B and then falls back to O (Fig. 4.5). Can this be considered a motion in a straight line?**

✔ ANSWER

Yes. The ball goes up and comes down along the same vertical straight line, so it is motion in a straight line (in both directions).

2 **Fill up Table 4.1. From Fig. 4.5: O = 0 cm, A = 40 cm, C = 80 cm, B = 140 cm (top). The ball goes O → A → B → C → O.**

✔ ANSWER

S. No.	Position	Total distance travelled from O	Displacement from O
1.	O	0 cm	0 cm
2.	A	40 cm	40 cm upward
3.	B	140 cm	140 cm upward
4.	C	140 + 60 = 200 cm	80 cm upward
5.	O	200 + 80 = 280 cm	0 cm

3 **Which is true for displacement?**

(i) It is never zero.

(ii) Its magnitude can be greater than the total distance travelled.

(iii) Its magnitude is less than or equal to the total distance travelled.

(iv) Its magnitude is less than the total distance travelled in all cases.

✔ ANSWER

(iii). Displacement became zero at the end (so (i) is wrong); it was equal to the distance while going up (so (iv) is wrong) and never more than the distance (so (ii) is wrong).

⭐ Pause and Ponder · 4.1.2 Distance travelled and displacement

1 **In the example of an athlete running back and forth on a straight track (Fig. 4.4), when will the displacement of the athlete be zero? What will be the total distance travelled in that case?**

✔ ANSWER

Displacement is **zero when she comes back to the starting point O**.

If she runs from O to A (100 m) and back to O (100 m), total distance = 100 m + 100 m = **200 m**.

2 **Fuel used up in a vehicle depends on which of the following? (i) Total distance travelled (ii) Displacement. Justify your answer.**

✔ ANSWER

(i) Total distance travelled.

The engine burns fuel for every metre the vehicle actually moves, whatever the direction. A vehicle that goes out and comes back has zero displacement but still uses fuel.

3 **A ball rolls down an inclined track (Fig. 4.6): OA = 40 cm, AB = 10 cm, BC = 20 cm, CD = 30 cm.**

Is its motion a straight line motion? Can its motion from O to D be shown using a horizontal line as in Fig. 4.3? Are the total distance travelled and magnitude of displacement from O equal or different at A, B, C and D?

✔ ANSWER

Yes, it moves along a straight (slanting) line in one direction.

Yes, we can draw the track as a straight line with O as origin and mark A at 40 cm, B at 50 cm, C at 70 cm and D at 100 cm (the line only needs to be straight; it need not be horizontal in reality).

Position	Distance from O	Displacement from O
A	40 cm	40 cm

B	50 cm	50 cm
C	70 cm	70 cm
D	100 cm	100 cm

They are **equal** at every position, because the ball moves in one direction without turning back.

⭐ Pause and Ponder · 4.1.3 Average speed and average velocity

4 **During a family road trip, you drive 200 km north in three hours. Afterwards, you drive 200 km south in two hours. Find the average speed and average velocity for your entire trip.**

✔ ANSWER

Total distance = 200 km + 200 km = 400 km; total time = 3 h + 2 h = 5 h

Average speed = 400 km ÷ 5 h = **80 km h^{-1}**

Displacement = 200 km north − 200 km south = 0 (back to start)

Average velocity = 0 ÷ 5 h = **0 km h^{-1}**

5 **Under what condition(s) is the**
(i) magnitude of average velocity of an object equal to its average speed?
(ii) magnitude of average velocity of an object zero while its average speed is not zero?

✔ ANSWER

(i) When the object moves in a **straight line in one direction** without turning back (then distance = magnitude of displacement).

(ii) When the object moves and **comes back to its starting point** (displacement = 0, but distance is not 0), e.g. one full round of a circular track.

🧪 Activity 4.2: Let us calculate

1–2 **Look up the time various cars take to go from 0 to 100 km h^{-1}, record it in Table 4.2 and calculate the magnitude of average acceleration for each car.**

💡 SAMPLE ANSWER

100 km h^{-1} = 100 × 1000 m ÷ 3600 s ≈ 27.8 m s^{-1}; initial velocity = 0

Average acceleration = 27.8 m s^{-1} ÷ time taken. Example values:

Car	Time (0–100 km h^{-1})	Average acceleration
Car 1	8 s	27.8 ÷ 8 ≈ **3.5 m s^{-2}**

Car 2	12 s	27.8 ÷ 12 ≈ **2.3 m s^{-2}**
Car 3	16 s	27.8 ÷ 16 ≈ **1.7 m s^{-2}**

Conclusion: the less time a car takes, the greater its acceleration. (Use the real times you find.)

Activity 4.3: Let us plot a graph

1–7 **Plot the position-time graph for the data of Table 4.3 (0 s → 0 m, 1 s → 20 m, ... 6 s → 120 m) using X-axis: 5 divisions = 1 s and Y-axis: 5 divisions = 20 m.**

✔ ANSWER

Time on the X-axis, position on the Y-axis; plot (0, 0), (1 s, 20 m), (2 s, 40 m) ... (6 s, 120 m).

Observation: all points lie on a **straight line** through the origin.

Conclusion: the vehicle covers equal distances (20 m) in equal times, so it moves with **constant velocity** (20 m s^{-1}).

Activity 4.4: Let us calculate

1–4 **In the position-time graph (Fig. 4.14), draw triangle ABC on part AB. What do BC and CA represent? Find the average velocity.**

✔ ANSWER

BC = change in position ($s_2 - s_1$); CA = change in time ($t_2 - t_1$).

From the graph: s_1 = 40 m at t_1 = 2 s; s_2 = 80 m at t_2 = 4 s

v = BC ÷ CA = (80 m − 40 m) ÷ (4 s − 2 s) = 40 m ÷ 2 s = **20 m s^{-1}**

Conclusion: the **slope** of a position-time graph gives the velocity.

⭐ Bridging Science and Society · 4.3 Kinematic equations

? **Can you now understand why it is important to maintain a safe distance from the vehicle moving ahead, and how this distance needs to be adjusted given your initial velocity?**

✔ ANSWER

Yes. A vehicle cannot stop at once: it moves during the driver's reaction time and then while braking.

Braking distance = $u^2/(2a)$ grows with the **square** of the speed (Example 4.8: 28.1 m at 54 km h^{-1} but 112.5 m at 108 km h^{-1}).

So at higher speed, on wet roads or with worn brakes, we must keep a **larger** gap.

🧪 Activity 4.5: Let us investigate · 4.4.1 Uniform circular motion

3–5 **A marble moves along the inner boundary of a ring. Predict, then lift the ring while the**

marble is moving. Does the marble continue moving in a circle?

✔ ANSWER

Observation: as soon as the ring is lifted, the marble **moves in a straight line**; it does not go on in a circle.

Conclusion: it keeps moving in the direction it had at that instant, i.e. along the **tangent** to the circle. The ring was needed to keep changing its direction.

⭐ Revise, Reflect, Refine

1 **My father went to a shop 250 m from home on a straight road. He came home to take a cloth bag, went to the shop again, bought provisions and came back home. How much was the total distance travelled by him? What was his displacement from home?**

✔ ANSWER

He went home → shop → home → shop → home = 4 trips of 250 m

Total distance = 4 × 250 m = **1000 m (1 km)**

He ended at home, so displacement = **0 m**

2 **A student runs from the ground floor to the fourth floor and then comes down to the classroom on the second floor. Each floor is 3 m high. Find: (i) the total vertical distance**

travelled, (ii) their displacement from the starting point.

✔ ANSWER

Ground → 4th floor = 4 × 3 m = 12 m up; 4th → 2nd floor = 2 × 3 m = 6 m down

(i) Total vertical distance = 12 m + 6 m = **18 m**

(ii) Final position = 2 × 3 m = 6 m above ground, so displacement = **6 m upward**

3 **A girl's scooter speedometer reading is constant. Is it possible for her scooter to be accelerating and if so, how?**

✔ ANSWER

Yes. Velocity has both magnitude and direction. If she is going around a curve or a circular turn, the direction of velocity changes even though the speed stays the same.

A change in direction means a change in velocity, so the scooter is accelerating (like uniform circular motion).

4 **A car starts from rest and its velocity reaches 24 m s^{-1} in 6 s. Find the average acceleration and the distance travelled in these 6 s.**

✔ ANSWER

Given: u = 0, v = 24 m s^{-1}, t = 6 s

a = (v − u) ÷ t = (24 − 0) ÷ 6 = **4 m s^{-2}**

$s = ut + \frac{1}{2}at^2 = 0 + \frac{1}{2} \times 4 \times 6^2 = 2 \times 36 =$ **72 m**

5 **A motorbike moving with initial velocity 28 m s^{-1} and constant acceleration stops after travelling 98 m. Find the acceleration of the motorbike and the time taken to come to a stop.**

✔ ANSWER

Given: $u = 28$ m s^{-1}, $v = 0$, $s = 98$ m

$v^2 = u^2 + 2as \rightarrow 0 = 28^2 + 2 \times a \times 98 \rightarrow 0 = 784 + 196a$

$a = -784 \div 196 =$ **-4 m s^{-2}** (opposite to the direction of motion)

$v = u + at \rightarrow 0 = 28 + (-4)t \rightarrow t = 28 \div 4 =$ **7 s**

6 **Fig. 4.27 shows position-time graphs of objects A and B moving along parallel tracks in the same direction (both are straight lines; B starts ahead, A's line is steeper and crosses B's line at about 5 s). Do A and B ever have equal velocity? Justify.**

✔ ANSWER

No. Both graphs are straight lines, so each object moves with a constant velocity.

The line of A is steeper (greater slope) than that of B, so A's velocity is always greater than B's.

Where the lines cross (about 5 s), A and B have the **same position**, not the same velocity.

7 **Fig. 4.28: A (straight line) and B (curve) start at the same position at 0 s and reach the same position at 10 s, moving in a straight line. Choose the correct option(s).**

(i) The average velocity of both over 10 s is equal since they have the same initial and final positions.

(ii) The average speeds of both over 10 s are equal since both cover equal distance in equal time.

(iii) The average speed of A is lower than that of B since it covers a shorter distance than B in 10 s.

(iv) The average speed of A is greater than that of B since B's speed is lower than A's in some segments.

✔ ANSWER

(i) and (ii) are correct.

Same initial and final positions in the same 10 s → same displacement → same average velocity.

Both move in one direction (the graphs never go down), so distance = displacement for both → equal distances in 10 s → equal average speeds. (iii) and (iv) are wrong: average speed depends only on total distance and total time.

8 **A truck driver driving at 54 km h^{-1} sees a speed limit sign of 40 km h^{-1}. He slows down to 36 km h^{-1} in 36 s. What distance did he travel during this time? (Constant acceleration.)**

✔ ANSWER

u = 54 km h^{-1} = 15 m s^{-1}, v = 36 km h^{-1} = 10 m s^{-1}, t = 36 s

a = (v − u) ÷ t = (10 − 15) ÷ 36 = −5/36 m s^{-2} (≈ −0.14 m s^{-2})

s = ut + ½at^2 = 15 × 36 + ½ × (−5/36) × 36^2 = 540 − 90

s = 450 m

9 **A car starts from rest and accelerates uniformly to 20 m s^{-1} in 5 s. It then travels at 20 m s^{-1} for 10 s and finally brakes uniformly to stop in 6 s. Find the total distance travelled.**

✔ ANSWER

Part 1 (speeding up): s_1 = ½(u + v)t = ½ × (0 + 20) × 5 = 50 m

Part 2 (constant velocity): s_2 = 20 × 10 = 200 m

Part 3 (braking): s_3 = ½ × (20 + 0) × 6 = 60 m

Total distance = 50 + 200 + 60 = **310 m**

10 **A bus is travelling at 36 km h^{-1} when the driver sees an obstacle 30 m ahead. He takes**

0.5 s to react. After braking, velocity reduces with constant acceleration of 2.5 m s^{-2}. Will the bus stop before reaching the obstacle?

✔ ANSWER

u = 36 km h^{-1} = 10 m s^{-1}

Distance during reaction time = 10 m s^{-1} × 0.5 s = 5 m

Braking distance: $v^2 = u^2 + 2as$ → $0 = 10^2 + 2 \times (-2.5) \times s$ → s = 100 ÷ 5 = 20 m

Total stopping distance = 5 m + 20 m = 25 m

25 m < 30 m, so **yes, the bus stops 5 m before the obstacle.**

11 **A student said, "The Earth moves around the Sun". In this context, discuss whether an object kept on the Earth can be considered to be at rest.**

💡 SAMPLE ANSWER

Rest and motion depend on the **reference point** we choose.

With respect to the Earth (e.g. the floor or walls of a room), the object's position does not change, so it is **at rest**.

With respect to the Sun, the object moves along with the Earth, so it is **in motion**.

So the object is at rest and in motion at the same time, depending on the reference point.

12 **Fig. 4.30 (velocity-time graph of a cyclist): velocity rises from 0 to 3 m s^{-1} in 0–20 s, stays 3 m s^{-1} from 20 s to 100 s, then falls to 2 m s^{-1} at 120 s.**

Shade the areas showing displacement (i) while moving with constant velocity, (ii) when velocity is decreasing. Also calculate the displacement and average acceleration in the 120 s interval.

✔ ANSWER

(i) Shade the **rectangle** under the flat line from 20 s to 100 s (one colour): area = 3 m s^{-1} × 80 s = 240 m

(ii) Shade the **trapezium** under the falling line from 100 s to 120 s (another colour): area = ½ × (3 + 2) × 20 = 50 m

Triangle from 0 to 20 s: ½ × 20 × 3 = 30 m

Total displacement = 30 + 240 + 50 = **320 m**

Average acceleration = (v − u) ÷ t = (2 − 0) ÷ 120 = **1/60 m s^{-2} ≈ 0.017 m s^{-2}**

13 **Fig. 4.31 (velocity in km h^{-1} vs time in h, read from the graph): 7.0 from 0 to 0.5 h; rises to 7.5 at 1.5 h; 7.5 till 3 h; falls to 7.0 at 4.5 h and 6.5 at 5.5 h; 6.5 till 6.5 h.**

Estimate the distance she ran.

💡 SAMPLE ANSWER

Distance = area under the velocity-time graph (she runs in one direction).

0–0.5 h: 7.0 × 0.5 = 3.5 km

0.5–1.5 h: ½ × (7.0 + 7.5) × 1 = 7.25 km

1.5–3 h: 7.5 × 1.5 = 11.25 km

3–4.5 h: ½ × (7.5 + 7.0) × 1.5 = 10.875 km

4.5–5.5 h: ½ × (7.0 + 6.5) × 1 = 6.75 km

5.5–6.5 h: 6.5 × 1 = 6.5 km

Total ≈ **46 km** (check the readings on the graph in your book)

14 **A car moves with constant velocity 6 m s^{-1} for 2 minutes and then accelerates at 1 m s^{-2} for 6 s. Find its displacement in 2 min 6 s by drawing a velocity-time graph.**

✔ ANSWER

Graph: a flat line at 6 m s^{-1} from 0 to 120 s, then a straight line rising to v = 6 + 1 × 6 = 12 m s^{-1} at 126 s.

Area of rectangle (0–120 s) = 6 × 120 = 720 m

Area of trapezium (120–126 s) = ½ × (6 + 12) × 6 = 54 m

Displacement = 720 + 54 = **774 m**

15 **Cars A and B start from rest with constant acceleration. A reaches 5 m s^{-1} in 5 s; B reaches 3 m s^{-1} in 10 s. Plot both velocity-time graphs on the same graph and calculate the displacement in the two time intervals.**

✔ ANSWER

Acceleration: $a_A = 5 \div 5 = 1\ m\ s^{-2}$; $a_B = 3 \div 10 = 0.3\ m\ s^{-2}$

Car A: time	1 s	2 s	3 s	4 s	5 s
velocity ($m\ s^{-1}$)	1	2	3	4	5
Car B: time	2 s	4 s	6 s	8 s	10 s
velocity ($m\ s^{-1}$)	0.6	1.2	1.8	2.4	3.0

Both graphs are straight lines from the origin; A's line is steeper.

Displacement of A (0–5 s) = area of triangle = ½ × 5 × 5 = **12.5 m**

Displacement of B (0–10 s) = ½ × 10 × 3 = **15 m**

16 **Rohan studies from 6 PM to 7:30 PM. For the tip of the minute hand (length 7 cm) during this time, find its (i) distance travelled, (ii) displacement, (iii) speed, and (iv) velocity.**

✔ ANSWER

In 90 min the minute hand makes 1½ rounds; the tip goes from 12 back to 12 and then to 6.

(i) Distance = 1.5 × 2πr = 1.5 × 2 × 22/7 × 7 cm = **66 cm**

(ii) Displacement = diameter = 2 × 7 = **14 cm**, from the 12 mark towards the 6 mark (downward)

(iii) Speed = 66 cm ÷ 90 min ≈ **0.73 cm per minute** (≈ 0.012 $cm\ s^{-1}$); it is the same at every instant (uniform circular motion)

(iv) Velocity at any instant has the same magnitude (≈ 0.73 cm/min) but its direction is along the tangent and **keeps changing**.

Average velocity for the 90 min = 14 cm ÷ 90 min ≈ **0.16 cm per minute**, from 12 towards 6.

⭐ The Journey Beyond

1 **Spin a disc with numbers 1–12 written 7 cm from the centre and letters A–F written 4 cm from the centre. Why do the numbers fade while the letters stay visible? Are their speeds the same or different?**

✔ ANSWER

All points make one round in the same time T, but the numbers move on a bigger circle.

Speed = 2πr/T, so the numbers move 7/4 times faster than the letters. **Speeds are different.**

Being faster, the numbers blur and fade first; the slower letters remain visible.

2 **Use an accelerometer app (e.g. Phyphox, 'Accelerometer without g'). Note readings when the phone is (i) on an outstretched palm, (ii) on the floor. What do you observe?**

💡 SAMPLE ANSWER

(i) On the palm the readings keep changing slightly (small non-zero values), because our hand makes tiny involuntary movements.

(ii) On the floor the readings stay almost zero.

So real objects that look still may have small accelerations; perfect rest is rare.

3 **Using Eq. (4.4a) and (4.4b), derive $s = vt - \frac{1}{2}at^2$ and $s = \frac{1}{2}(u + v)t$. Derive the second one using the area of a trapezium.**

✔ ANSWER

$s = vt - \frac{1}{2}at^2$: from $v = u + at$, $u = v - at$. Put in $s = ut + \frac{1}{2}at^2$:

$s = (v - at)t + \frac{1}{2}at^2 = vt - at^2 + \frac{1}{2}at^2 =$ **$vt - \frac{1}{2}at^2$**

$s = \frac{1}{2}(u + v)t$: in the velocity-time graph (Fig. 4.19) the displacement is the area of trapezium OABD, whose parallel sides are AO = u and BD = v, and width OD = t.

Area of trapezium = ½ × (sum of parallel sides) × distance between them

$s =$ **$\frac{1}{2}(u + v)t$**

4 **Plot graphs for Table 4.4 using different X and Y scales and compare. Which scale is better and why?**

💡 SAMPLE ANSWER

With a small scale the curve looks flat and crowded; with a very large scale it does not fit the paper.

The better scale is the one that uses most of the graph paper and makes points easy to read (e.g. 5 divisions = 2 s, 5 divisions = 5 m). Graph apps choose such scales automatically.

5 **Talk to a motor mechanic about how stopping distance is affected by wet roads, worn-out tyres, higher mass, driving at night, fog, severe weather and driver reaction time; make safety posters and a skit.**

💡 **SAMPLE ANSWER**

All these factors **increase** the stopping distance:

- Wet roads, worn tyres, rain/snow: less grip, so brakes slow the vehicle less.
- Higher mass: brakes give less deceleration, so more distance.
- Night, fog, storms: hazards are seen late, so reaction distance increases.
- Longer reaction time: the vehicle travels farther before braking starts.

Poster idea: “Slow down and keep a safe distance – stopping takes longer than you think!”

🎉 **Chapter 4 done! Revise the answers once more tomorrow.**